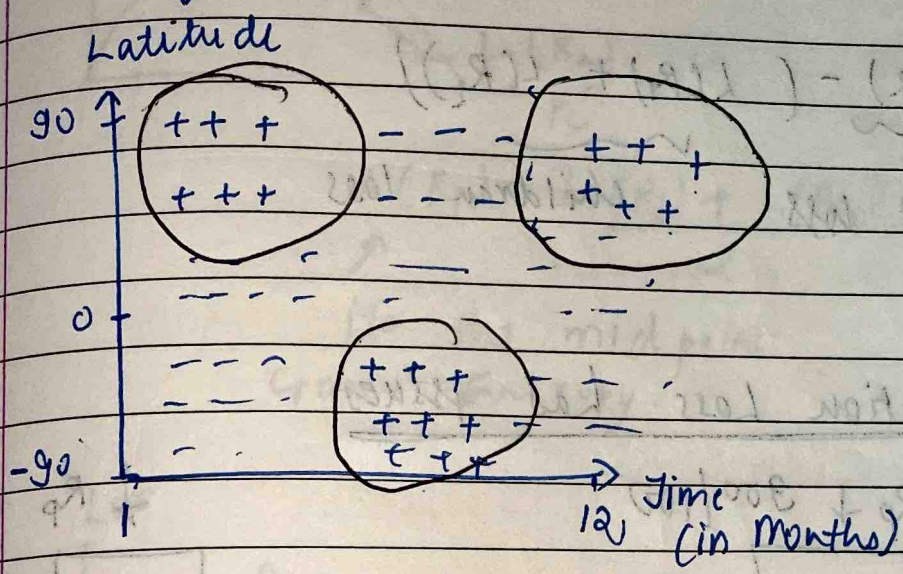
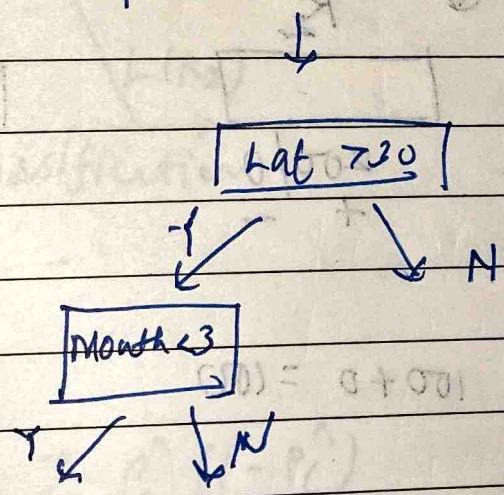


Lecture - 10 - Decision Trees & Ensemble methods

- Decision Trees
- Ensemble Methods
- Bagging
- Random Forests
- Boosting



Greedy, Top-Down, Recursive Partitioning



Region R_p , Looking for a split S_p

$$S_p(j, t) = \left(\begin{array}{l} \{x \mid x_j < t, x \in R_p\} \rightarrow R_1 \\ \{x \mid x_j \geq t, x \in R_p\} \rightarrow R_2 \end{array} \right)$$

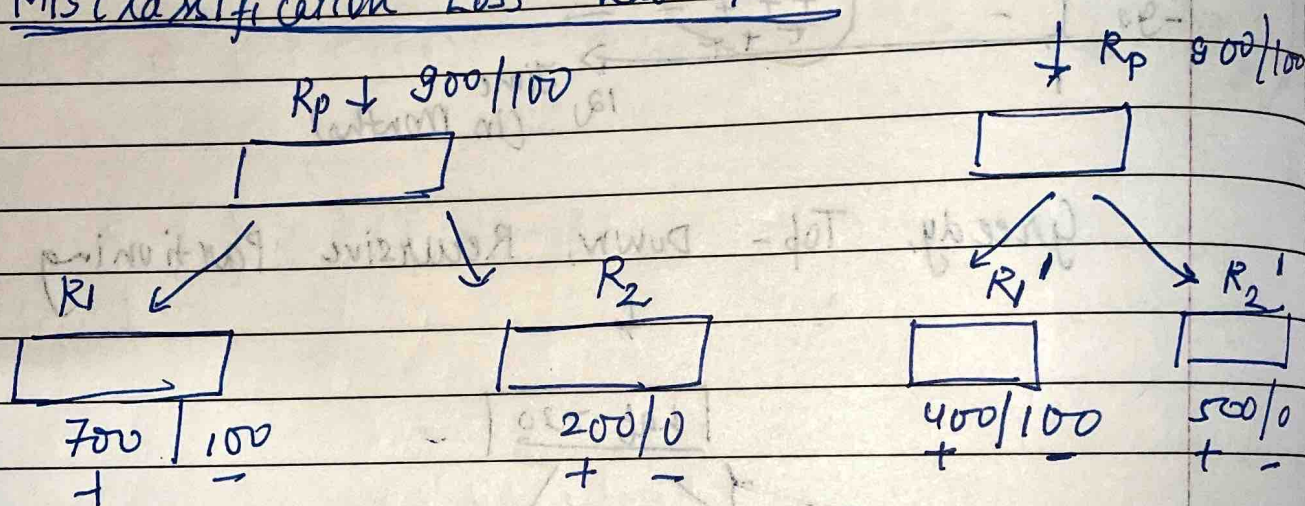
How to choose splits?

Define $L(R)$: Loss of R given C classes.
 define $p_c^{\wedge}$ to be the proportion of class c samples in R that are

$$L_{\text{misclass}} = 1 - \max_c p_c^{\wedge}$$

$$\max_{j,t} \min_{i,t} \underbrace{L(R)}_{\text{parent loss}} - (\underbrace{L(R_1)}_{\text{children loss}} + \underbrace{L(R_2)}_{\text{children loss}})$$

Misclassification Loss has issues



$$L(R_1) + L(R_2) = 100 + 0 = 100$$

$$L(R_1') + L(R_2')$$

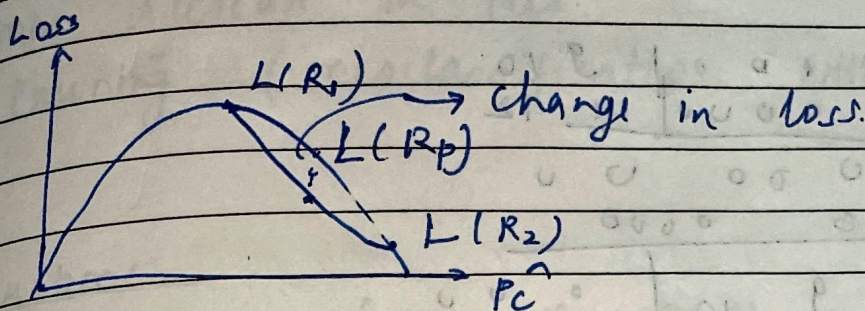
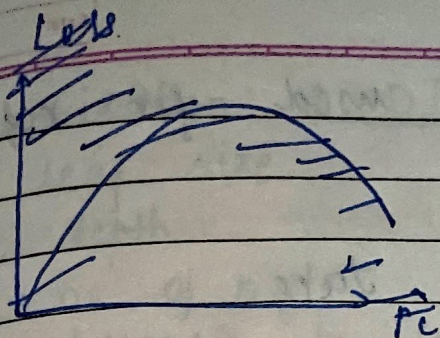
$$= 100 + 0$$

$$= 100$$

$$L(R_p) = 100$$

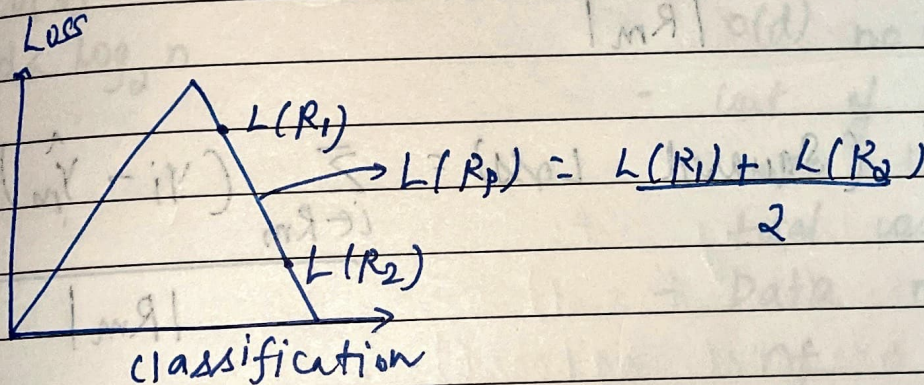
Instead, define cross-entropy loss

$$L_{\text{cross}} = \sum_c p_c^{\wedge} \log_2 p_c^{\wedge}$$



$$L(R_p) = L(R_1) + L(R_2)$$

↑
its the midpoint
Cross-Entropy.



Gini-Loss

$$\sum_c p_c^{\hat{}} (1 - p_c^{\hat{}})$$

Decision Trees can be used for regression problems also.

They are called Regression Trees

Lat

Northern Hemisphere	7 8	1 0 0	9 10
	9 10	0 0 0 0	11 12
Equator	0 0	0 0	0 0
	0 0	0 0 0 0	
Southern Hemisphere	0 0	9 10	0 0 0
	0 0	15 5	0 0 0

Inches of snowfall

Predict R_m

$$\hat{y}_m = \frac{\sum_{i \in R_m} y_i}{|R_m|}$$

$$L_{\text{squared}} (\text{Squared Loss}) = \frac{\sum_{i \in R_m} (y_i - \hat{y}_m)^2}{|R_m|}$$

Decision Trees can be used for categorical variables also.

q categories

2^q possible splits.

Regularization of DT

- 1) min leaf size
- 2) max depth
- 3) max no. of nodes
- 4) min decrease in loss.
- 5) Pruning (misclassification with validation set).

Runtime

Suppose we have n examples, f features & depth d .

Test time

Train side

$$O(d)$$

$$d \leq \log_2 n$$

- Each point is part of $O(d)$ nodes

- Cost of point at each node is $O(f)$, so total cost is $O(nfd)$

- Data matrix is of size nf .

The downside of decision trees is that it has no additive structure.

Recap of Decision Trees

- | | |
|--|--|
| <ul style="list-style-type: none"> ⊕ Easy to explain ⊕ Interpretable ⊕ Categorical variables ⊕ Fast. | <ul style="list-style-type: none"> ⊖ High Variance ⊖ Bad at additive ⊖ Low predictive accuracy. |
|--|--|

Ensembling

Take X_i 's which are random variables (RV) that are independent identically distributed (iid)

$$\text{Var}(X_i) = \sigma^2 \quad \text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_i X_i\right) = \frac{\sigma^2}{n}$$

Dropping the independence assumption, so now X_i 's are id
 X_i 's are correlated by ρ

$$\text{Var}(\bar{X}) = \rho\sigma^2 + \frac{1-\rho}{n}\sigma^2$$

Ways to ensemble

- (i) different algorithms
- (ii) different training sets
- (iii) Bagging (Random Forest)
- (iv) Boosting (Ada boost, Xgboost).

Bagging - Bootstrap aggregation. Have a true population P

Let training set be S .
 $S \sim P$

Assume $P = S$

Bootstrap samples $Z \sim S$

Bootstrap Samples $Z_1, Z_2, \dots, Z_M$

train model G_m on $\bar{Z}_m$

$$G(m) = \frac{\sum_{m=1}^M G(x)}{m}$$

Bias - Variance Analysis

$$\text{Var}(x) = f(1-f)^2 + \frac{(1-f)^2}{M}$$

$$\text{Var}(x) = f\sigma^2 + \frac{1-f}{M}\sigma^2$$

Bootstrapping is driving down σ^2

More $M \rightarrow$ less variance, bias gets slightly increased because of random subsampling.

DTs + Bagging

DTs have high variance, low bias, so this is an ideal fit for bagging.

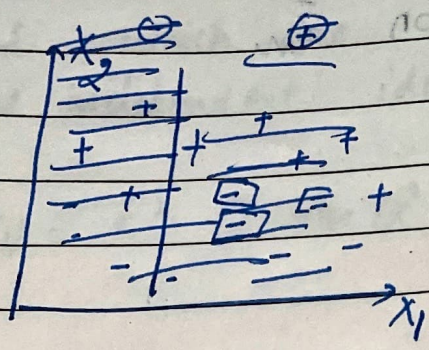
Random Forests

At each split, consider a fraction of your total features.

Decreases σ^2

Boosting

We decrease bias, also it is additive



$$f(x) = f_1(x) + f_2(x) + \dots + f_M(x)$$

Boosting is combining weak classifiers

More $M \rightarrow$ less variance, bias

DTE + Boosting

DTE has high variance, low bias, so this is ideal fit for boosting

Random Forests

at each split, consider a fraction of features